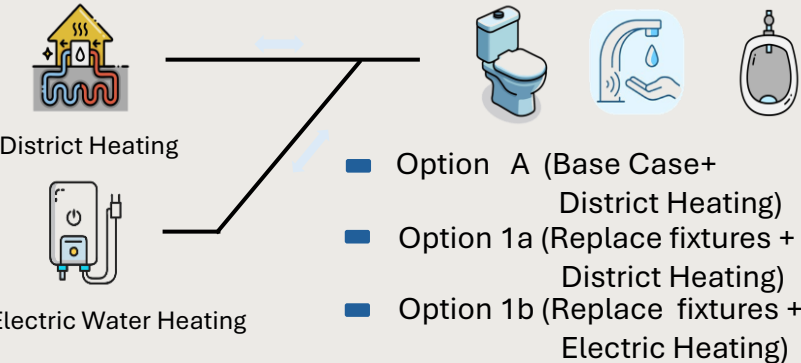




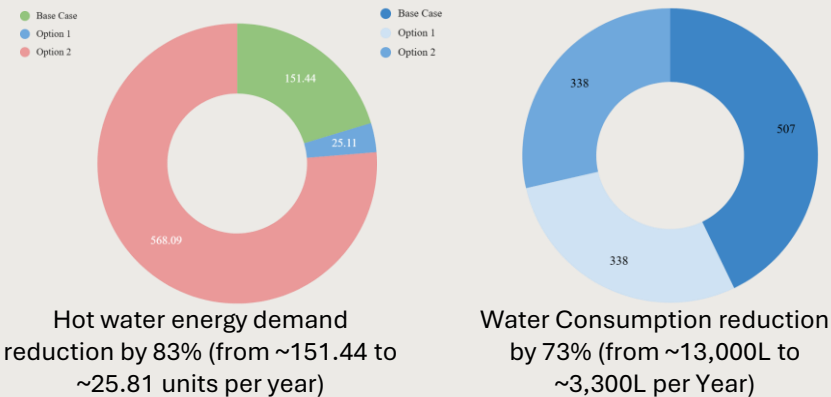
Introduction

This study applies an integrated Life Cycle Sustainability Assessment (LCSA) to compare a building's base case with two renovation measures. The methodology combines Life Cycle Assessment (LCA) and Life Cycle Costing (LCC), synthesized via a custom weighting system. A sensitivity analysis tests assumptions, with results presented in a comparative poster for clear decision support.

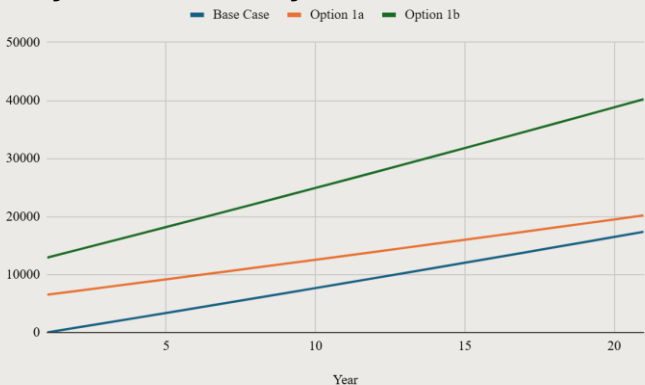
Methodology



Usage

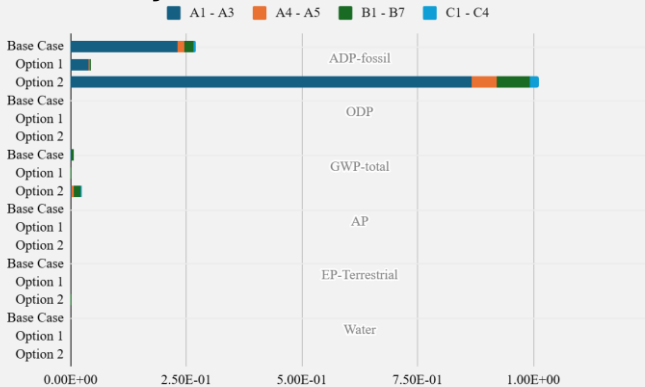


Life Cycle Cost: Payback Time

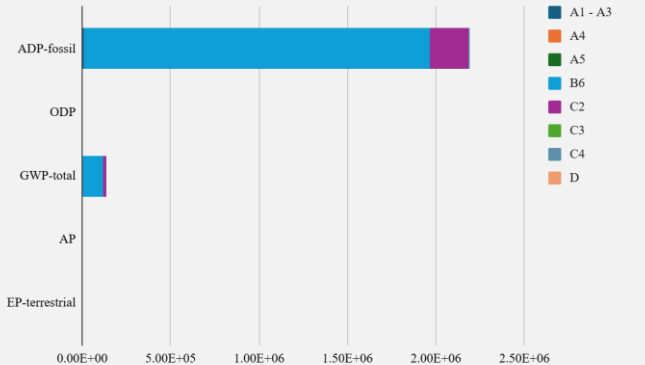


Op1 compared to the Base Case is approximately 34.2 years, which exceeds the analysis period, while Op2 does not achieve payback.

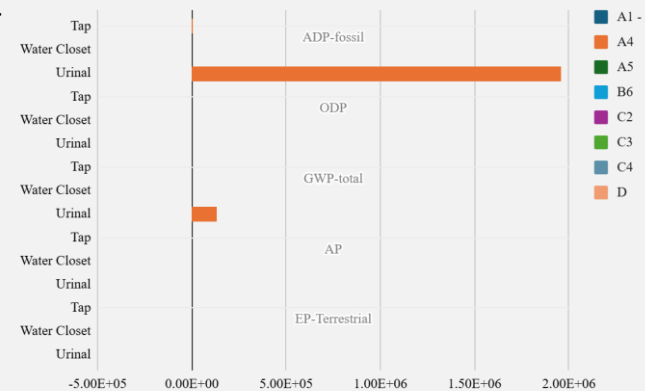
Integrated Life Cycle Assessment



The base case shows the highest ADP fossil impact. District heating with fixture change (Option 1a) significantly reduces fossil depletion, while the electric boiler with fixture change (Option 1b) performs best overall, minimizing impacts across most categories such as GWP, AP, and EP terrestrial.



The base case (Option 1a) shows the highest ADP fossil impact. District heating with fixture change (Option 1b) reduces fossil depletion notably. The electric water heater with fixture change (Option 3) achieves the lowest overall impact across most categories, indicating improved environmental performance.



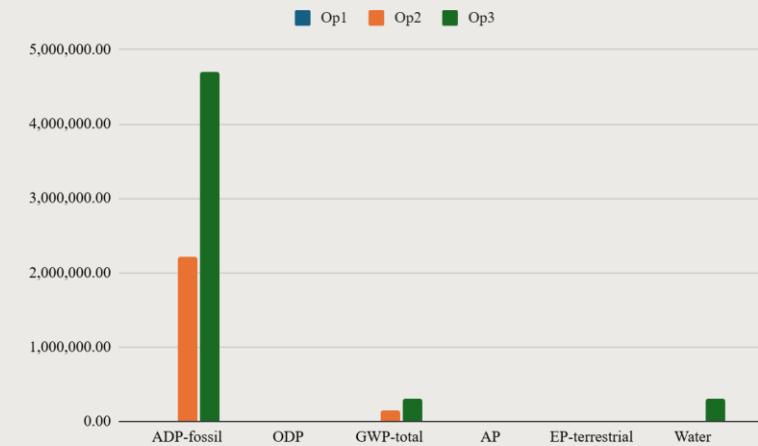
For fixtures, the ADP fossil category dominates, mainly from stage A4, showing high fossil resource depletion. Other categories, such as ODP, GWP total, AP, and EP terrestrial, have minimal impacts, indicating transportation significantly influences overall environmental performance.

References

- The Netherlands Organisation for Applied Scientific Research (TNO) and Delft University of Technology.[<https://www.ecocostsvalue.com/ecocosts/data/>]
- Guinée, J. B., Heijungs, R., Huppes, G., Zamagni, A., Masoni, P., Buonamici, R., & Rydberg, T. (2016). Life Cycle Assessment: Past, Present, and Future. *Environmental Science & Technology*, 50(13), 6874–6881.

Dutch shadow-cost

Impact Category	Unit	Shadow Cost (EUR/unit)	Price Conversion (SEK)
ADP-fossil	MJ	0.1	1.13
ODP	kg CFC-11	43	485.9
GWP	kg CO ₂ -eq	0.133	1.5029
AP	kg SO ₂ -eq	10.7	120.91
EP	kg PO ₄ ²⁻ -eq	8.55	96.615
Water	Liter	0.0016	0.01808



This LCA uses the Dutch shadow-cost method to monetize impacts and compare options. Results show Op3 has the highest external costs, dominated by ADP-fossil and with notable Water and GWP, Op2 is lower but still fossil-driven, Op1 is minimal across categories, indicating the best environmental performance and priority direction for design.

Results:

Fixture replacement reduces water demand by 83% and consumption by 73%. LCA shows Option 1b (electric heating) has lowest environmental impacts across life cycle stages. Shadow costing reveals Op3 has highest external costs from ADP-fossil and water use, Op2 moderate fossil impacts, while Op1 achieves minimal monetized environmental costs.

Conclusion

- Option A (Base Case):** No upfront costs but the highest energy use, water consumption, and environmental impacts, leading to the greatest shadow costs and unsustainable long-term performance.
- Option 1a (Fixtures + District Heating):** Reduces hot water energy by 83% and water use by 73%, but payback exceeds 34 years and shadow costs remain moderate.
- Option 1b (Fixtures + Electric Heating):** Achieves minimal environmental and shadow costs with top resource efficiency, yet initial investment precludes payback within analysis period.

- de Bruijn, H., & Huijbregts, M. A. J. (2016). Handbook on the Life Cycle Assessment of Heating Technologies. CE Delft. Retrieved from CE Delft Shadow Prices Handbook.
- Goedkoop, M., Heijungs, R., Huijbregts, M., De Schryver, A., Struijs, J., & van Zelm, R. (2013). ReCiPe 2008: A Life Cycle Impact Assessment Method Which Comprises Harmonised Category Indicators at the Midpoint and the Endpoint Level. *Report 1: Characterisation*. Netherlands Institute for Public Health and the Environment (RIVM).